Logic Puzzles

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CPS721: Artificial Intelligence I

Learning Objectives

Students should be able to

1. Encode logic puzzles using Prolog

2. Use hidden variables to better capture constraints

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The Puzzle

Chris, Sandy, and Pat discover they all have distinct occupations and play distinct musical instruments.

Figure out who plays which instrument using the following clues:

1. Chris is married to the doctor
2. The lawyer plays the piano
3. Chris is not the engineer
4. Sandy is a patient of the violinist
5. One of them plays the flute

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Variables and Domains

- Several ways to encode this problem

- We will do the following

Variables are `Doc, Law, Eng, Piano, Violin, Flute`

- Will set `chris, sandy, pat` as the domain of these variables

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Implicit Constraints

- Without looking at the clues, what do we know about the values of `Doc, Law, Eng`?

- What about `Piano, Violin, Flute`?

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Generating Valid Assignments

```
person(chris).person(sandy).person(pat).
distinct(X, Y, Z) :-
```

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Implicit Constraints

- If X is married to Y, does that tell us anything about X and Y?

- If X is a patient of Y, what does that tell us about X and Y?

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Solving the Puzzle

```
solve([Doc, Law, Eng, Piano, Violin, Flute]) :-
      distinct(Doc, Law, Eng),
      distinct(Piano, Violin, Flute),
```

% Chris is married to doctor

% Lawyer plays the piano

% Engineer is not Chris

% Sandy is patient of violinist

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Changing the Puzzle

Old Clues

1. Chris is married to the doctor
2. The lawyer plays the piano
3. ~~Chris is not the engineer~~
4. Sandy is a patient of the violinist
5. One of them plays the flute

New Clues

1. Chris is married to the doctor
2. The lawyer plays the piano
3. Pat is not married to the engineer
4. Sandy is a patient of the violinist
5. One of them plays the flute

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Changing the Puzzle

Can express 3 as

"Pat is not the engineer"

But missing out on possible connections to 1

New Clues

1. Chris is married to the doctor
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Using Hidden Variables

- Will add hidden variables to capture spouse relationships
  - Not part of answer, but useful for propagating the effect of constraints
  - Like carry values in cryptarithmetic

- Notice there are three possible (general) arrangements
  - Chris and Sandy are married
  - Chris and Pat are married
  - Sandy and Pat are married
  - Nobody is married (to each other)

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Adding Spouse Information

- Create predicate to enforce possible spouse relationships

```
spouses(ChrisSpouse, SandySpouse, PatSpouse)
```

In KB, put the following

```
spouses(sandy, chris, none).
spouses(pat, none, chris).
spouses(none, pat, sandy).
spouses(none, none, none).
```

Interpret as “these are all possible”

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Solving the Puzzle

```
solve([Doc, Law, Eng, Piano, Violin, Flute]) :-
      distinct(Doc, Law, Eng),
      distinct(Piano, Violin, Flute),
      spouses(ChrisSpouse, SandySpouse, PatSpouse),
```

% Chris is married to doctor

% Lawyer plays the piano

% Pat is not married to the engineer

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